Bi 2021 Kidney RCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

5/8 samples (natural-break ≥160 cells) intermix on one continuous manifold; no per-patient islands (41 detached cells removed). Diff-sample kNN 0.46 = 0.67 of the mixing ceiling; 1 component.

Gate 2 — module geography → GO

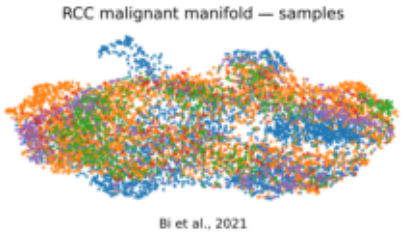
Non-branching: Gast/pSom co-vary from a shared origin (no opposing arms); Crest is the inverse terminal gradient (Crest–pSom r = −0.65; Crest vs pseudotime = +0.71). pSom globally strongest.

Gate 3 — hub mutual exclusivity → NA

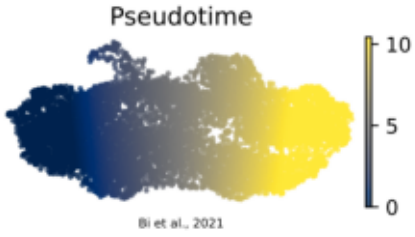
No bifurcation and no Gast/pSom hub — the branching-only cross-repression test does not apply.

STEP FIGURES

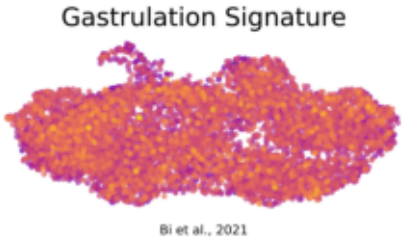
Manifold — samples



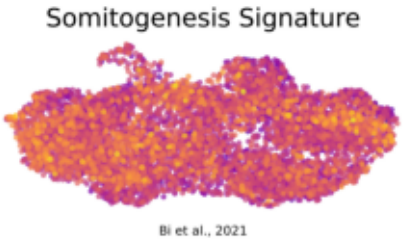
Pseudotime



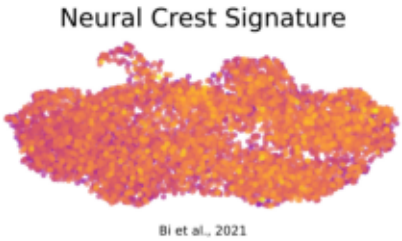
Gastrulation



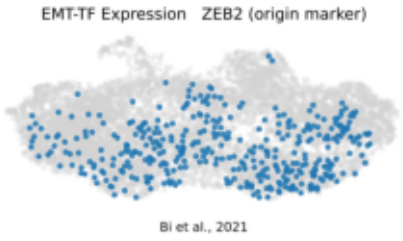
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Kidney renal-cell carcinoma (Bi et al., 2021), 10x — 8 patients / 34,326 cells (master scope); malignant only: 8,040 load, 7,822 across 5 samples enter the trajectory.

Verdict.
Pattern reproduces (non-branching). A single embryonic axis runs from a Gast/pSom-high origin to a Neural-crest-high terminus — the programs are present but not resolved into competing arms, so there is no hub and Gate 3 is NA. EMT-TFs: canonical factors all decline toward the terminus; only ZEB2 clears the detection floor (19%) and marks the mesenchymal/gastrulation origin — no bipolar EMT-TF split.

Parameter choices

Min cells/sample	≥160 (natural break, keeps 5/8)	**Integration**	Harmony on sample, θ=10
Island removal	UMAP radius-graph (−41 cells)	**Trajectory**	scFates curve N=20 (non-branching)
Root node	tip 0 = Gast/pSom origin	**Modules**	score_genes, union control pool
n_pcs / spread	20 / 1.3	**Overdispersion**	scFates find_overdispersed